



Multilinguality at Common Crawl

Improving Language Coverage for the Largest Open Web Corpus

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Common Crawl – A Brief Introduction

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WMDQS

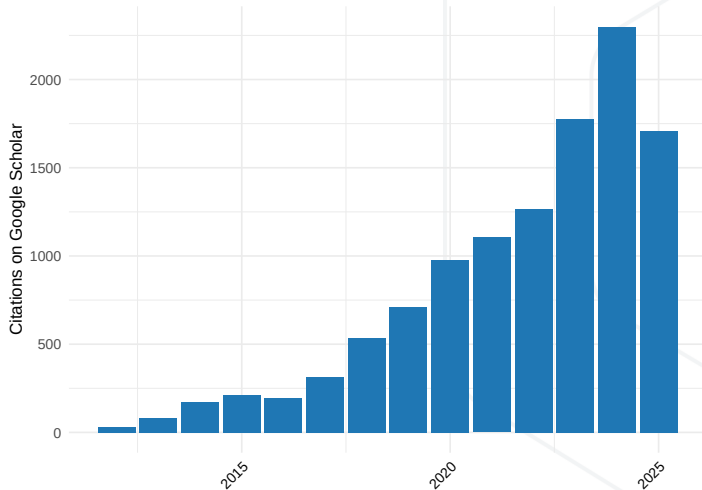
CommonLID Dataset

Summary

About Common Crawl

- We're a non-profit making web data accessible to programmers and data scientists
- Free archive of the public web since 2007, founded by Gil Elbaz
- Hosted as Open Dataset on Amazon Web Services [1, 2]
- Cited in over 10,000 research papers

Papers Citing Common Crawl (through 9/25)



Common Crawl Use Cases

- Search indexes
- Sociopolitical research
- Linguistic analyses
- Internet security research
- Economic research
- AI/ML training
- A lot more...

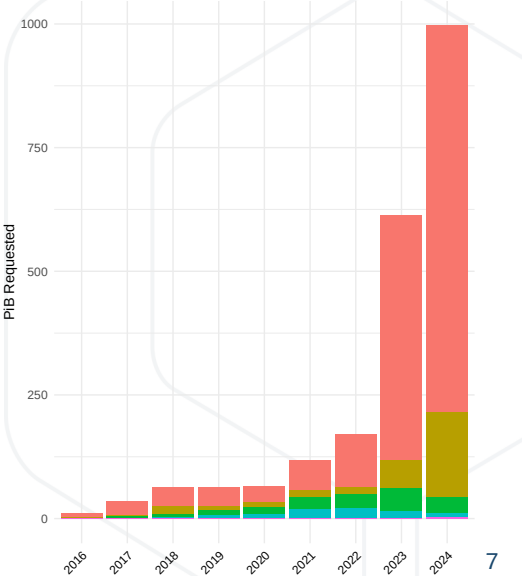
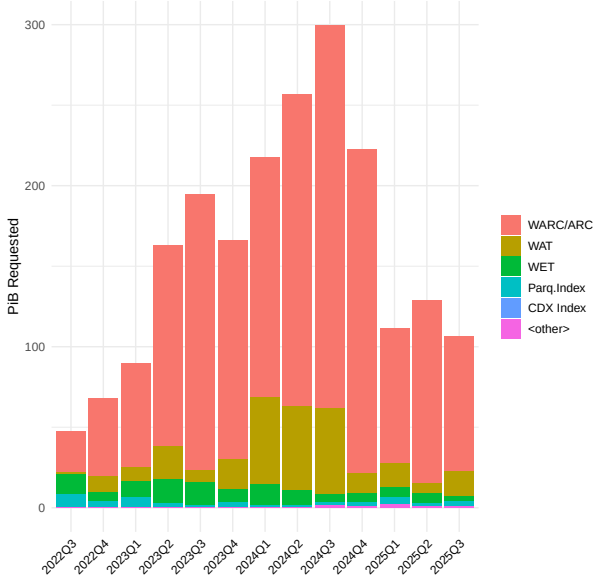
Our Mission

- Make high-quality public web crawl data (once limited to large search engines) available to everyone
- Support people, governments, and organizations make data-driven decisions and tackle global issues
- Build an open knowledge ecosystem that encourages sharing and innovation
- Support understanding and preservation of language, community, and culture

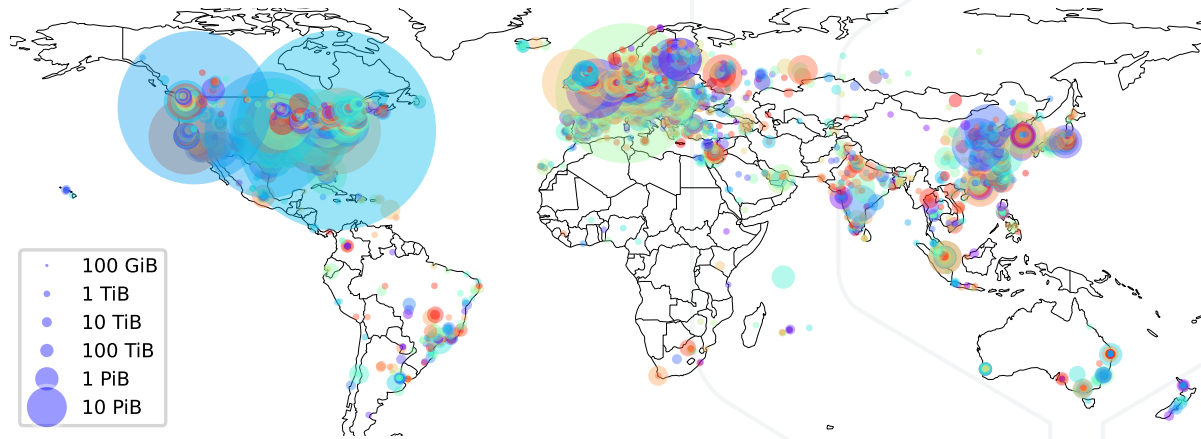
Data Overview

- Over 300 billion web pages spanning 17 years (2008 – 2025)
- Around 2.5 billion pages added each month
- More than 100 crawl archives released to date
- 10.1 PiB of data (Sept 2025)
- Additional data products: web graph, host index, etc.

Data Downloads



Downloads by Geographical Location



Data Collection – A Look Back In Time

Four phases of data collection using different

- Crawler implementations
- Approaches to find and sample (prioritize) seeds and URLs
- Page revisit policies

	crawler	seeds / link prioritization	revisit policy
2008–2009	Nutch	list based	
2012	in-house	page rank	
2013 – 2016	Nutch	seed donations, list based	
2017 – now	Nutch	harmonic centrality	30% new, 70% revisits based on page relevance

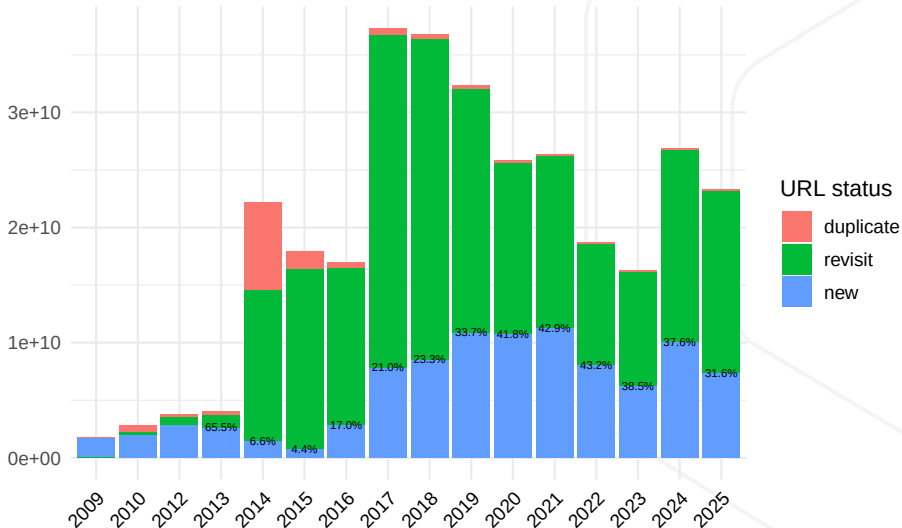
What Is Representative?

Aspects of representativity:

- Breadth: coverage of unique domains (web sites)
- Depth: per-site coverage
- Freshness (new content)
- Amount of (near-)duplicates (per crawl and over multiple crawls)
- Regional coverage (top-level domains, content languages)
- Content quality
- Applicable for a given data use case?

Size and Freshness

Number of Page Captures



Crawler Politeness

- Crawl slowly
 - Further slow down (exponential backoff) if a site responds with errors
- Respect robots.txt rules (Robots Exclusion Protocol – [3]) and
- URI-level metatags (`<meta name=robots value=nofollow>`) [4, 5]
- CCBot identifies itself
 - User-agent string and contact information sent along with requests
 - Crawling from fixed list of IP addresses, publicly announced and verifiable via reverse DNS

Common Crawl – A Brief Introduction

Data Pipelines

Prevalence of Web Data in LLMs

Anatomy of a Data Pipeline

Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

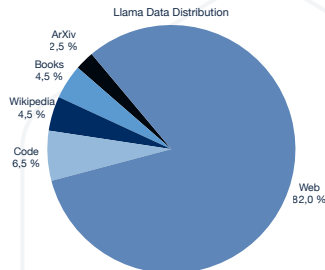
WMDQS

CommonLID Dataset

Summary

Prevalence of Web Data in LLMs Today(ish)

- In practice datasets are not balanced
- Web data is always the cheapest and easiest to get
- Web data is diverse, but definitely not balanced or representative of all the language range.
- Web data always contains unwanted content (fiction, bias, propaganda).
- Programming language code (source code) is becoming ubiquitous in the data mix
- Public domain books and encyclopedic data is also common, but availability varies greatly between languages.



LLaMA Data	
Source	Proportion
Web	82%
Code	6.5%
Wikipedia	4.5%
Books	4.5%
ArXiv	2.5%

Anatomy of a Data Pipeline

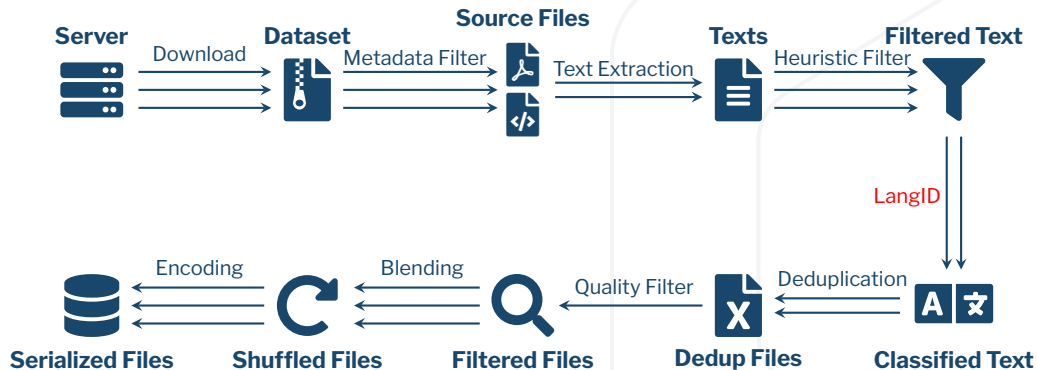


Figure 1: A diagram of a common pre-processing pipeline. Arrows represent parallel processes; the more arrows there are, the more parallelizable the specific task is.

A Survey of Pre-Processing Pipelines

- OSCAR (First use of LangID threshold for filtering) [6, 7]
- CCNet (Uses KenLM models [8] trained on Wikipedia) [9]
- mC4 (Uses word blocklists) [10]
- Refined Web (Uses Trafilatura for text extraction) [11]
- DataTrove (Uses Trafilatura and aggregates filters) [12]
- Dolma (Works on deduplication) [13]
- NeMo-Curator (Combines many heuristics and filters) [14]
- HPLT (Uses custom pipeline and has bilingual data) [15]
- MADLAD (Uses custom LangID and focuses on Multilinguality) [16]
- GlotCC (Uses Ungoliant [7] plus a new LangID model) [17]
- Community OSCAR (KenLM models trained on Adult Content) [18]
- And many others!

Common Crawl – A Brief Introduction

Data Pipelines

Multilingual Data Pipelines

Multilinguality is Hard

Plurality of Crawl is English

Language Identification Isn't Perfect

Current CCF LID: CLD2

Common Crawl's Multilingual Initiatives

WMDQS

CommonLID Dataset

Summary

Multilinguality is Hard – Even with Web Data!

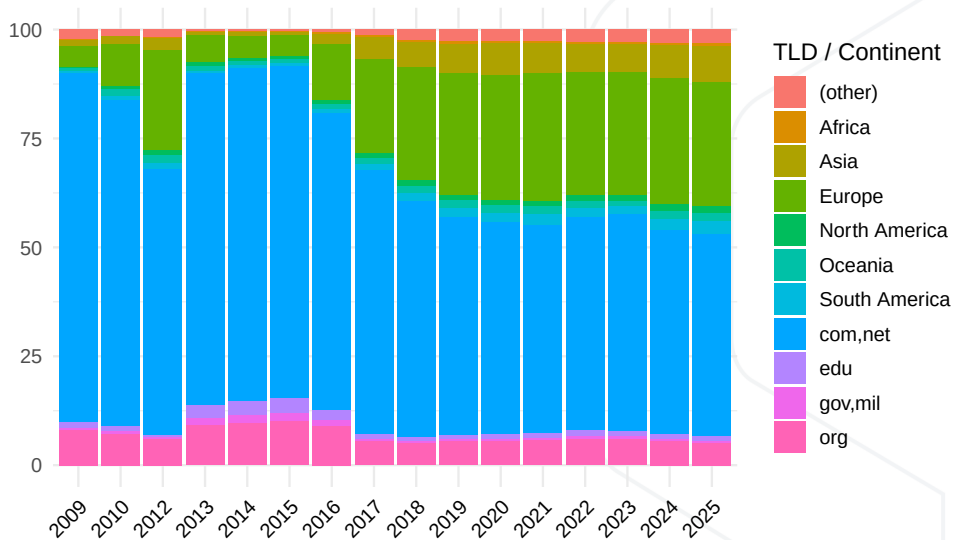
- The hope: lots of multilingual data on the web
- The reality: current pipelines struggle with multilinguality
 - **Plurality of crawl is English**
 - **Language identification isn't perfect**
 - Extraction and preprocessing not fully multilingual

Language Coverage Skews Towards English

	2018	2019	2020	2021	2022	2023	2024	2025
<other>	6.47	7.31	7.88	7.59	7.51	7.84	7.90	7.91
<unknown>	3.32	2.62	2.35	2.53	2.80	2.84	3.00	2.91
ara	0.76	0.59	0.56	0.60	0.63	0.64	0.66	0.67
ces	1.03	1.04	1.05	1.06	1.03	1.09	1.04	1.02
deu	5.15	5.47	5.57	5.63	5.60	5.79	5.36	5.57
eng	43.96	43.84	43.20	44.98	46.64	45.70	44.40	44.07
fas	0.66	0.59	0.57	0.63	0.63	0.67	0.71	0.72
fra	4.53	4.56	4.53	4.46	4.50	4.64	4.31	4.31
ind	0.75	0.74	0.76	0.83	0.75	0.82	1.01	1.09
ita	2.06	2.31	2.38	2.42	2.48	2.69	2.53	2.31
jpn	5.47	4.81	4.78	4.66	4.68	4.83	5.03	5.09
kor	0.59	0.69	0.76	0.66	0.65	0.67	0.71	0.77
nld	1.50	1.74	1.79	1.80	1.93	2.09	1.85	1.81
pol	1.68	1.70	1.68	1.62	1.62	1.70	1.80	1.75
por	1.99	2.07	2.16	2.15	1.78	1.27	2.14	2.26
rus	9.27	7.41	7.11	7.07	5.90	5.84	6.02	5.97
spa	4.18	4.16	4.25	4.33	4.36	4.58	4.56	4.46
tur	0.90	0.87	0.91	1.00	0.86	0.84	1.18	1.15
vie	0.75	0.74	0.80	0.92	0.92	1.04	1.01	1.03
zho	4.98	6.76	6.91	5.07	4.73	4.41	4.75	5.12

Percentage of yearly crawls in selected languages, as identified by CLD2 Source: [19]

Top-Level Domain Coverage Skews Western

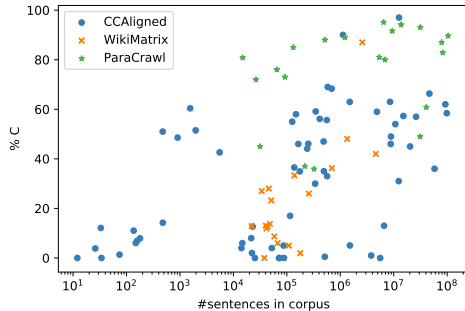
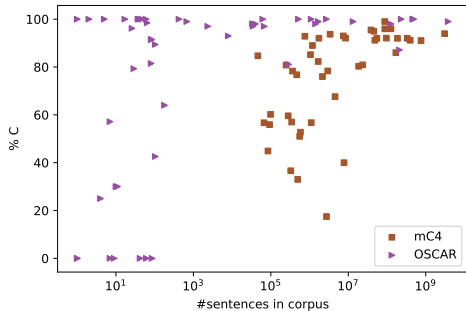


Human Evaluation Shows Big Web LID Problems

		Parallel			Monolingual	
		CCAligned	ParaCrawl v7.1	WikiMatrix	OSCAR	mC4
#langs audited / total		65 / 119	21 / 38	20 / 78	51 / 166	48 / 108
%langs audited		54.62%	55.26%	25.64%	30.72%	44.44%
#sents audited / total		8037 / 907M	2214 / 521M	1997 / 95M	3517 / 8.4B	5314 / 8.5B
%sents audited		0.00089%	0.00043%	0.00211%	0.00004%	0.00006%
macro	C	29.25%	76.14%	23.74%	87.21%	72.40%
	X	29.46%	19.17%	68.18%	-	-
	WL	9.44%	3.43%	6.08%	6.26%	15.98%
	NL	31.42%	1.13%	1.60%	6.54%	11.40%
	offensive	0.01%	0.00%	0.00%	0.14%	0.06%
	porn	5.30%	0.63%	0.00%	0.48%	0.36%
micro	C	53.52%	83.00%	50.58%	98.72%	92.66%
	X	32.25%	15.27%	47.10%	-	-
	WL	3.60%	1.04%	1.35%	0.52%	2.33%
	NL	10.53%	0.69%	0.94%	0.75%	5.01%
	offensive	0.00%	0.00%	0.00%	0.18%	0.03%
	porn	2.86%	0.33%	0.00%	1.63%	0.08%
#langs =0% C		7	0	1	7	0
#langs <50% C		44	4	19	11	9
#langs >50% NL		13	0	0	7	1
#langs >50% WL		1	0	0	3	4

Taken from [20]

High-Resource $\neq$ High Quality



Taken from [20]

CCF's Internal Eval (2018) => We Use CLD2

Test Set	Languages	Accuracy				CPU Time cumul.
		dsl2014	europarl	twitter	tatoeba	
Docs		12600	21000	187461	19457	
MiB		3.3	3.4	18.8	0.9	
Languages		10	21	137	143	
cld2 [21]	160	.879	.989	.751	.792	4
cld2 (Java bind) [21, 22, 23]	83	.875	.986	.743	.606	13
cld2 (Java bind) [21, 22, 23]	160	.880	.990	.758	.792	14
cld2 Polyglot (Py bind) [24]	160	.879	.989	.748	.792	14
cld3 [25]	101	.830	.991	.595	.630	29
fasttext [26]	176	.880	.991	.753	.787	4
idNet [27]	463	.682	.997	.721	.680	3:07:07
langid (Python) [28]	97	.790	.992	.695	.604	19:19
langid (C) [29]	97	.790	.992	.695	.604	7
langid (Java) [30]	97	.790	.992	.695	.604	20
optimaize [31]	71	.740	.969	.534	.443	1:25
shuyo-cybozu [32]	52	.761	.993	.620	.420	56

About CLD2

- Two variants: 83 or 160 supported languages
- Detects up to 3 languages per document
- Plain text or HTML input, valid UTF-8
- Quadgram Naïve Bayesian classifier with support of
 - Writing system / Unicode script
 - Ignoring web-specific character sequences (*http, copyright*)
 - Top-level domain of URL
 - Language codes in HTML and HTTP metadata
 - Character encoding (BIG5, KOI8-R)
- C++ library, trained on a proprietary corpus; no support for re-training

Language Annotations in Common Crawl Data

- WARC metadata record

```
charset-detected: EUC-KR
languages-cld2: {
  "reliable":true,
  "text-bytes":2404,
  "languages":[
    {"code":"ko", "code-iso-639-3":"kor",
      "text-covered":0.97, "score":3735.0, "name":"Korean"},
    {"code":"en", "code-iso-639-3":"eng",
      "text-covered":0.02, "score":1264.0, "name":"ENGLISH"}]}
```

- WAT (same as in WARC metadata record)

- WET

WARC-Identified-Content-Language: kor,eng

- URL indexes: kor,eng (up to 3 language codes)

Common Crawl – A Brief Introduction

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Multilingual Data Pipelines

Common Crawl's Multilingual Initiatives

Improving Language Coverage and Balance

Initiative 1: The Web-Languages Project

Initiative 2: Improving LID for the Web

WMDQS

CommonLID Dataset

Summary

Improving Language Coverage and Balance

Two main goals of current efforts:

- **Expand coverage** of crawl for languages other than English (LOTE)
- **Improve language identification** for web: more languages, better fidelity

Initiative 1: The Web-Languages Project

- Crowd-sourced effort to improve crawling of low-resource languages
- Language, Community, Culture
- Show the crawler high-quality links to “steer” into the right direction
- Hopefully, this will influence the harmonic centrality ranks over time

(<https://github.com/commoncrawl/web-languages>)

The Web-Languages Project

The screenshot displays the GitHub interface for the `web-languages` repository. At the top, the repository name `web-languages` is shown with a 'Public' badge. Navigation links for 'Unwatch', 'Fork', and 'Star' are visible. Below the repository name, a table lists the contents of the repository:

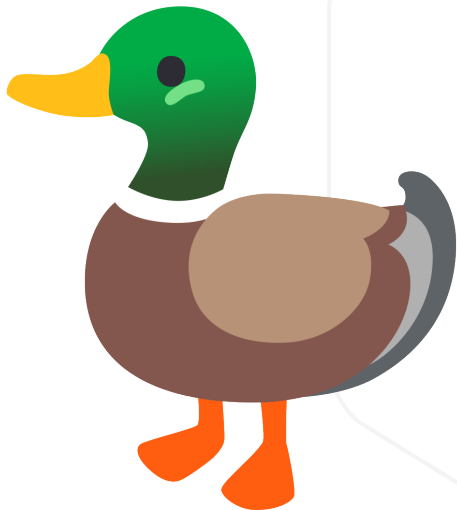
File/Folder	Description	Last Commit
<code>constructed</code>	Update volapük.md	9 months ago
<code>extinct</code>	feat: add wikipedia links for all languages	last year
<code>historical</code>	Update latin.md	2 months ago
<code>living</code>	Update iranian_persian.md	last week
<code>special</code>	feat: add wikipedia links for all languages	last year
<code>README.md</code>	Update README.md	3 months ago

The `README` file is selected and displayed below the table. It features the Common Crawl logo and the text 'LANGUAGE COMMUNITY CULTURE'. The title 'Web Languages Project' is prominently displayed, followed by a welcome message: 'Welcome! This is a crowd-sourced effort to improve crawling of low-resource languages. This dataset is public.'

On the right side of the repository page, the 'About' section provides context: 'Crowd-sourced lists of urls to help Common Crawl crawl under-resourced languages. See <https://github.com/commoncrawl/web-languages-code/> for the code'. It also includes links to the `commoncrawl.org/` website and tags for `crawling`, `language-detection`, and `dataset`. A sidebar on the right lists repository statistics: 58 stars, 7 watching, 73 forks, and a link to 'Report repository'. A 'Contributors' section shows 61 contributors with their avatars and a link to '+ 47 contributors'.

Figure 2: <https://github.com/commoncrawl/web-languages>

Initiative 2: Improving Language Identification for the Web



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The LangID Project

Why LangID

The Annotation Platform

The Incentives to Annotate

The Organizations Behind the Effort

The Hackathons

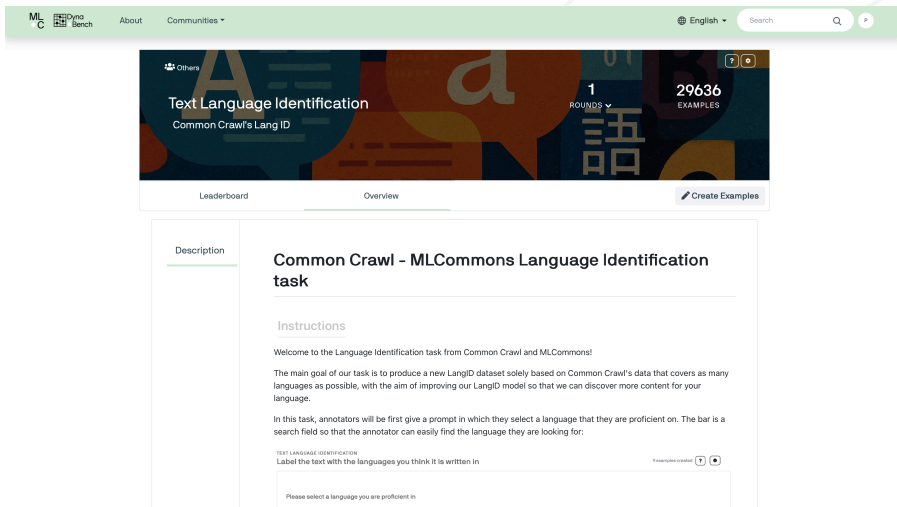
WMDQS

The Results

CommonLID Dataset

Summary

The LangID Project



The screenshot shows the MLCommons website interface for the LangID project. At the top, there is a navigation bar with the MLCommons logo, 'About', and 'Communities' links. A search bar is located on the right side of the navigation bar. Below the navigation bar, there is a large banner for the 'Text Language Identification' task. The banner features a colorful background with various characters and symbols. It includes the text 'Text Language Identification' and 'Common Crawl's Lang ID'. On the right side of the banner, there are statistics: '1 ROUNDS' and '29636 EXAMPLES'. Below the banner, there are two tabs: 'Leaderboard' and 'Overview'. The 'Overview' tab is currently selected. To the right of the tabs, there is a button labeled 'Create Examples'. Below the tabs, there is a section titled 'Description'. Under 'Description', there is a heading 'Common Crawl - MLCommons Language Identification task'. Below this heading, there is a section titled 'Instructions'. The 'Instructions' section contains the following text: 'Welcome to the Language Identification task from Common Crawl and MLCommons! The main goal of our task is to produce a new LangID dataset solely based on Common Crawl's data that covers as many languages as possible, with the aim of improving our LangID model so that we can discover more content for your language. In this task, annotators will be first give a prompt in which they select a language that they are proficient on. The bar is a search field so that the annotator can easily find the language they are looking for:'. Below the text, there is a form with the label 'TEXT LANGUAGE IDENTIFICATION' and the instruction 'Label the text with the languages you think it is written in'. There is a search bar and a button labeled 'Examples created' with a plus icon. Below the form, there is a text input field with the placeholder text 'Please select a language you are proficient in'.

MLCommons Dyna Bench About Communities

English Search

Others

Text Language Identification

Common Crawl's Lang ID

1 ROUNDS

29636 EXAMPLES

Leaderboard Overview

Create Examples

Description

Common Crawl - MLCommons Language Identification task

Instructions

Welcome to the Language Identification task from Common Crawl and MLCommons!

The main goal of our task is to produce a new LangID dataset solely based on Common Crawl's data that covers as many languages as possible, with the aim of improving our LangID model so that we can discover more content for your language.

In this task, annotators will be first give a prompt in which they select a language that they are proficient on. The bar is a search field so that the annotator can easily find the language they are looking for:

TEXT LANGUAGE IDENTIFICATION

Label the text with the languages you think it is written in

Examples created

Please select a language you are proficient in

Figure 3: Now closed

Why LangID

- Existing models are old and unmaintained
- LangID models keep being used as filters but nobody studies the impact of this
- High coverage models remain largely under-evaluated due to lack of data
- The question of generalization across domains is under-studied
 - Most LangID datasets contain very clean data
- We want to expand the language coverage of Common Crawl and we need better signals
- LangID models need to accurately classify noisy web text
- Crowd-sourced data annotation task to collect manual annotations of web data

The Annotation Platform

TEXT LANGUAGE IDENTIFICATION

Label the text with the languages you think it is written in

9 examples created ? 0

Please select a language you are proficient in

Select a language

Select

Figure 4: A screenshot of the annotation platform interface. The participant selects the language they want to annotate for.

The Annotation Platform

TEXT LANGUAGE IDENTIFICATION

Label the text with the languages you think it is written in

43 examples created

To tag text in an additional language, select that language from the dropdown, then select the text. To undo a language tag, click anywhere in the selected text.

Select all text area

Spanish; Castilian

1: spa 2: eng 3: fra

ESTUDIO DE LA EXPRESIÓN Y FUNCIÓN DE RECEPTORES DE APOLIPOPROTEÍNA E DURANTE LA DIFERENCIACIÓN DE UN MODELO NEURONAL

Resumen | Convocatoria | Responsable

Resumen

Las Entidades Neurodegenerativas tales como la enfermedad de Parkinson (EP) y enfermedad de Alzheimer (EA) constituyen las dos primeras entidades degenerativas del SNC, y por tanto son parte frecuente de la consulta especializada de Neurología. En los últimos años se han realizado muchos esfuerzos para determinar los factores tanto genéticos como ambientales que pueden llevar al estado patológico. Asimismo, los trastornos del desarrollo del sistema nervioso central son preocupación constante en la clínica. Ambos grupos de entidades, al parecer confluyen en una vía común que está muy relacionada con la señalización dependiente de los lípidos. El objetivo de este proyecto es esclarecer el papel de la proteína Apolipoproteína E (APOE) y sus receptores en el balance muerte/supervivencia celular estudiando sus efectos en un modelo neuronal. La primera fase del proyecto consiste en observar los niveles de expresión de cuatro de las proteínas que actúan como receptores para APOE, entre ellas LDLR (Low Density Lipoprotein Receptor), LRP1 (Lipoprotein Receptor Related Protein-1), VLDLR (Very Low Density Lipoprotein Receptor) y APOER2 (Apolipoprotein E Receptor 2) durante el periodo de diferenciación de la línea celular CAD (Cath.a differentiated). En la segunda fase, se agregará proteína APOE recombinante al medio de cultivo para analizar sus consecuencias en el número y viabilidad de las células en un contexto neurotóxico inducido por ceramida; y posteriormente se analizará sus efectos en vías de señalización cinasa de tirosina, específicamente la vía PI3K/AKT junto con sus componentes corriente abajo, así como la activación de vías de muerte celular como las caspasas.

Convocatoria

Nombre de la convocatoria: FACULTAD DE MEDICINA 2008 - CONVOCATORIA PARA EL ESTÍMULO A LA INVESTIGACIÓN A TRAVÉS DE PROYECTOS Y ENFOQUES ESTRATÉGICOS, DE PRIORIDADES E INTERDISCIPLINARIOS

Modalidad: Modalidad I: Apoyo a Grupos de Investigación de la Facultad de Medicina, a través de proyectos de investigación - Grupos Categorizados por COLCIENCIAS como C y los demás grupos de investigación de la Facultad de Medicina.

Optional annotations: Choose one or more of the following tags for the document (only if they apply).

☐ Sexual The text contains explicit sexual or adult content.

☐ Toxic The snippet contains offensive or harmful content.

☐ Non-Standard Orthography The snippet contains variations in spelling, punctuation, and written conventions that deviate from established language norms.

☐ Boilerplate The text snippet contains linguistic content, but most of the content (more than 50%) is part of a template, form, standard alert text, a menu, etc.

Submit

Skip and load a new text

Figure 5: The participant has highlighted all text as Spanish.

The Annotation Platform

TEXT LANGUAGE IDENTIFICATION

Label the text with the languages you think it is written in

43 examples created ? ⓘ

To tag text in an additional language, select that language from the dropdown, then select the text. To undo a language tag, click anywhere in the selected text.

Select all text area

English ▼ 1: spa 2: eng 3: fra

Esta sección trata los aspectos relacionados con la jugabilidad de las partidas, como el uso de determinado software, la manera de pilotar tu nave en diferentes situaciones, y consejos para gestionar tu imperio. Los juegos están organizados en secciones, así que asegúrate de que el juego para el que estás mirando la ayuda está seleccionado en la barra de la izquierda.

Esta sección contiene preguntas que no pertenecen a una categoría determinada o que son de interés general. »»

[EN] My gameplay question is not listed here. – what do I do? eng

[ES] Puedo aterrizar en planetas spa

[EN] My reputation with the Paranid is Enemy of Priest Duke 18%, and when I kill enemies it changes to 19%. Is this correct? eng

— Optional annotations: Choose one or more of the following tags for the document (only if they apply).

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Submit Skip and load a new text

Figure 6: A screenshot of the annotation platform interface. The participant has highlighted the English and Spanish text in the extract in different colours.

The Annotation Platform

TEXT LANGUAGE IDENTIFICATION

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43 examples created ? ⓘ

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☐ Sexual The text contains explicit sexual or adult content.

☐ Toxic The snippet contains offensive or harmful content.

☐ Non-Standard Orthography The snippet contains variations in spelling, punctuation, and written conventions that deviate from established language norms.

☐ Boilerplate The text snippet contains linguistic content, but most of the content (more than 50%) is part of a template, form, standard alert text, a menu, etc.

Submit Skip and load a new text

Figure 7: A screenshot of the annotation platform interface. The participant has highlighted the English and Spanish text in the extract in different colours.

The Incentives to Annotate

- Better LangID performance for your language!
- Contributors who annotated at least 100 documents are invited to be authors on the dataset paper
- Encourage the development of open-source LangID models
- More and higher-quality (and noisy!) multilingual data, especially for low-resource languages

The Organizations Behind the Effort



Masakhane Hackathon

AfriCC

African Common Crawl Datasets



URL Collection

Browse about the curated collection of African cultural URLs



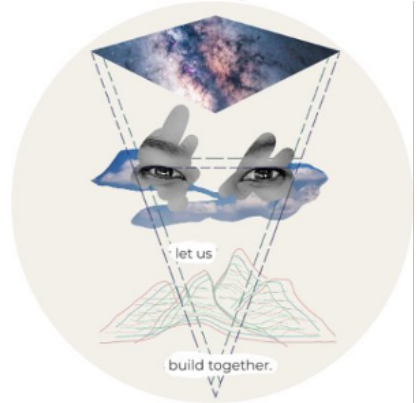
LID Annotation

Explore language identification annotations datasets across African languages



Contribute?

Interested in contributing to the AfriCC project? Join us!



Masakhane

Masakhane Hackathon

AfriCC LID Instances per Language

Comprehensive visualization of African language representation in the dataset

63

African Languages

9,698

Total Entries

154

Average per Language

All African Languages

Search languages...

Sort A-Z

Sort by Region

Sort by Instances

Language ↑	Region	Instances
Acoli	East Africa	1
Aja (Benin)	West Africa	3
Amharic	Horn of Africa	574
Antankarana Malagasy	Malagasy	1
Bambara	West Africa	15

zoom.us

SEACrowd Hackathon



The poster features a light gray background with a subtle geometric pattern of dots and lines. At the top, the SEACrowd logo is prominently displayed, with the 'SEA' part in a colorful, stylized font. Below the logo, the text 'X' indicates a collaboration. The logos for COMMON CRAWL, ML Commons, and EleutherAI are arranged horizontally. The main title 'Language Identification Hackathon' is written in a bold, red, sans-serif font. Two purple rounded rectangular boxes contain the 'When' and 'Where' information. The 'When' box lists the dates and times for different time zones. The 'Where' box includes a QR code for Google Meet. At the bottom, there are two more QR codes: one for joining the Discord server and another for the Annotation Platform, each with a small icon (Discord and a grid) next to it.

SEACrowd

X

COMMON CRAWL ML Commons ELEUTHERAI

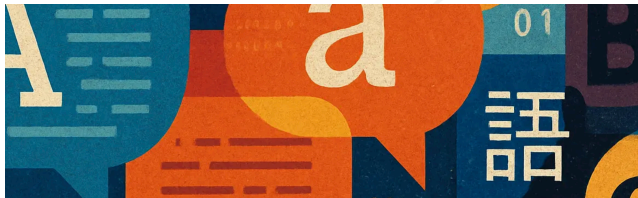
Language Identification Hackathon

When:
July 22, 2025
8-11am UTC -4 (US East Coast)
1-4pm UTC +1 (UK)
2-5pm UTC +2 (Central Europe)
7-10pm UTC +7 (Indochina Time)

Where:
Google Meet

Hear about future events
Join the Discord!

Annotation Platform



1st Workshop on Multilingual Data Quality Signals

Palais des Congrès
Montréal, Canada
10 October 2025

[Call for Papers](#) [Shared Task](#)

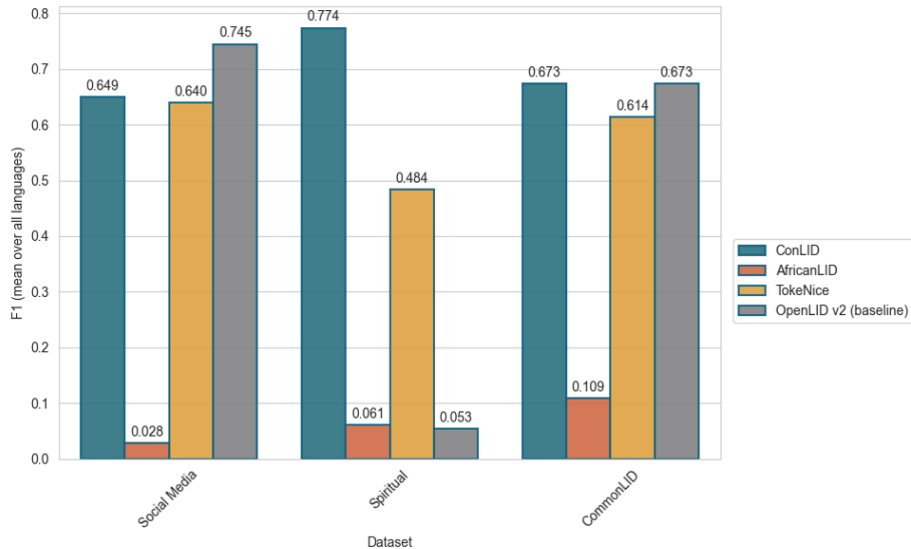
Recent research has shown that large language models (LLMs) not only need large quantities of data, but also need data of sufficient quality. Ensuring data quality is even more important in a multilingual setting, where the amount of acceptable training data in many languages is limited. Indeed, for many languages even the fundamental step of language identification remains a challenge, leading to unreliable language labels and thus noisy datasets for downstream language

Figure 8: <https://wmdqs.org>

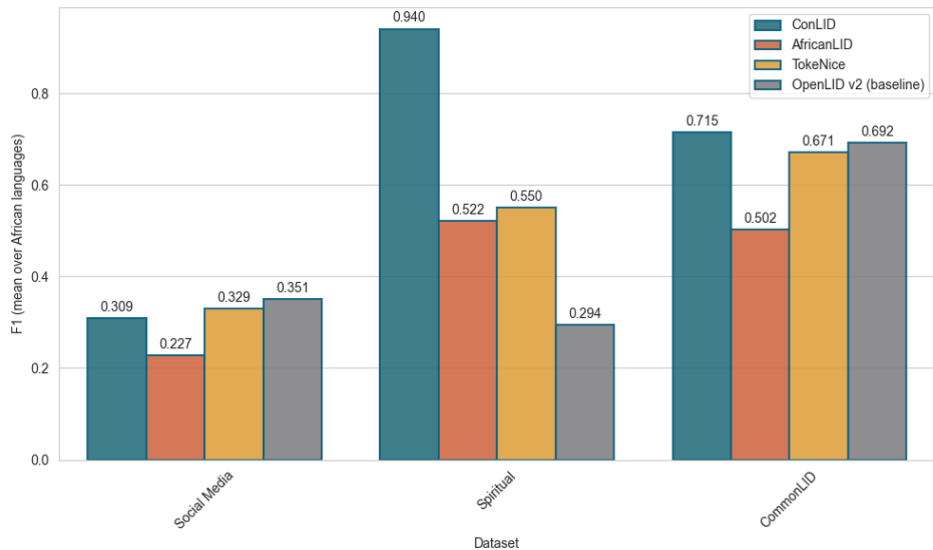
CommonLID v0.1 and a LangID Shared Taks

- CommonLID: 22K documents annotated or 200k annotated lines
 - 62 languages with >100 annotated lines
 - 42 languages with >500 annotated lines
 - 24 languages with >1000 annotated lines
 - 8 languages with >10000 annotated lines
- WMDQS LangID Shared Task
 - Evaluated on Dynabench
 - 18 different teams got into the shared task
 - 4 teams submitted models
 - 3 models ran on Dynabench
 - 3 Benchmarks were used: Social media dataset, spiritual text dataset and CommonLID

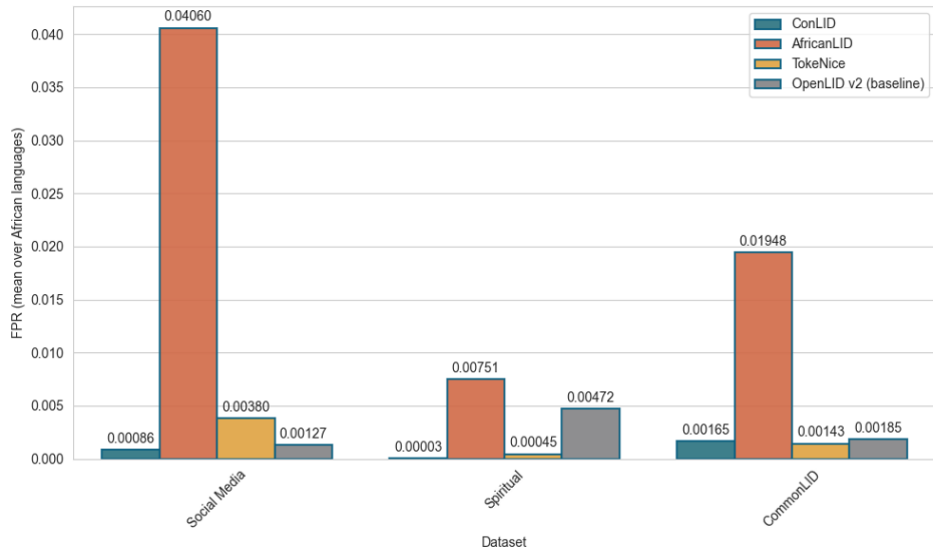
F1 Results: All Languages



F1 Results: African Languages



False-positive-rate: African Languages



Common Crawl – A Brief Introduction

Data Pipelines

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Common Crawl's Multilingual Initiatives

WMDQS

CommonLID Dataset

- Overview

- Creation

- Evaluation

Summary

Overview of CommonLID

Annotated data => LID evaluation dataset

- 373,230 lines
- 109 languages
- 78 languages with >100 lines
- 1-43,189 lines per class (mean 3424)
- 215.5 mean characters per line

All participants who annotated > 100 lines were offered co-authorship.

Geographic distribution of CommonLID

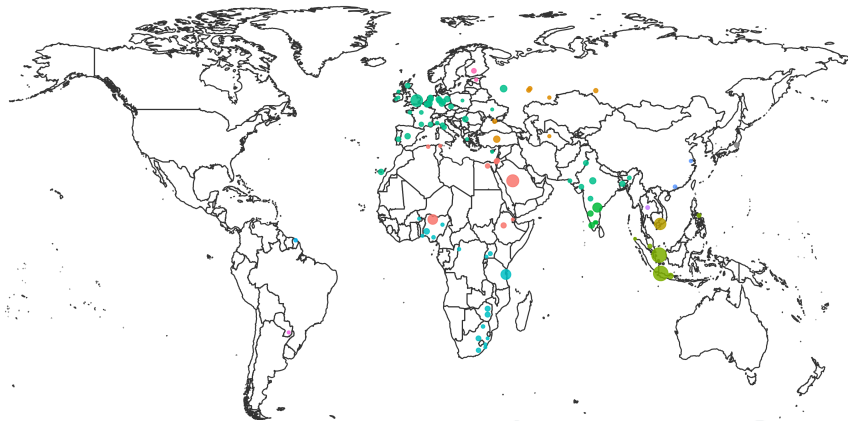


Figure 9: Illustrative geographical distribution of the language varieties in CommonLID. The dot size represents the number of annotated lines, and colour represents language family.

Creating CommonLID: Preprocessing

Converting to line-level annotations not trivial!

- Split on new lines
- Strip leading/trailing whitespace, de-duplicate line+language pairs
- Fuzzy match lines with multiple annotations

Creating CommonLID: Multiple Labels

Reconciling labels was a combination of automatic and manual processes

- Filter out short lines (< 10 characters)
- English contamination: `langid.py`, filter (most) lines $P(eng) > 0.7$
- Manual inspection of remaining multiple labels
 - Most remaining were clear mistakes => remove
 - Kept macro/micro language labels (both potentially valid)
 - Kept lines with multiple languages (none dominant)
- All languages audited (three spurious language classes removed)

Evaluating CommonLID

We evaluate 8 popular LID models using CommonLID + 5 evaluation datasets.

Fair comparison isn't easy:

- Label normalisation between eval sets and between model outputs (iso639-lang)
- Different models cover different languages: bigger not always better
- Web scale LID needs speed *and* fidelity!

Summary of Evaluated LID Models

Name	# langs.	Architecture	Data sources	Open data?
AfroLID [33]	517	Transformer	≈ 100M curated sents.	✗
CLD2 [21]	158	Naïve Bayes	Web pages (curated and scraped)	✗
fasttext [34]	218	FastText	“publicly available datasets”	✗
FUN-LangID [35]	1634	Common sub-strings	Web+Wikipedia+Bibles	✗
pyFranc [36]	414	Trigram distribution	UDHR	✓
gCLD3 [37]	99	Neural network	?	✗
GlottLID v3 [38]	1868	FastText	Curated open sources	✓
OpenLID-v2 [39]	193	FastText	Curated, audited open sources	✓

Table 1: Summary of LID models used.

Overlap in LID Model Coverage

	GlottLID	FUN-LangID	AfroLID	pyFranc	fasttext	OpenLID-v2	CLD2	gCLD3
GlottLID	1868							
FUN-LangID	1351	1549						
AfroLID	401	313	515					
pyFranc	362	312	93	410				
fasttext	202	180	49	166	210			
OpenLID-v2	190	157	47	156	180	193		
CLD2	128	153	30	121	114	104	158	
gCLD3	83	99	13	80	79	78	98	99

Table 2: Counts of mutual language coverage between LID models (models in descending order by coverage).

Macro-average F1 by model/evaluation set

	FLORES+ 209 languages		SmolSent 82 languages		UDHR-LID 418 languages		Bibles 1047 languages		Social Media 66 languages		CommonLID 109 languages	
	all	cov.	all	cov.	all	cov.	all	cov.	all	cov.	all	cov.
AL	15.9	67.9 ₍₄₉₎	42.3	69.4 ₍₅₀₎	15.2	67.6 ₍₉₄₎	12.1	69.8 ₍₁₈₁₎	1.5	98.0 ₍₁₎	9.2	43.5 ₍₂₃₎
C2	45.0	87.8 ₍₁₀₅₎	40.6	83.2 ₍₄₀₎	24.0	81.7 ₍₁₂₁₎	4.4	73.8 ₍₆₂₎	76.8	85.9 ₍₅₉₎	49.5	79.3 ₍₆₈₎
FT	80.3	92.2 ₍₁₈₂₎	48.1	83.8 ₍₄₇₎	27.0	68.2 ₍₁₆₆₎	3.3	43.8 ₍₇₉₎	83.6	84.9 ₍₆₅₎	49.3	72.6 ₍₇₄₎
FL	68.8	86.2 ₍₁₆₅₎	75.9	79.8 ₍₇₈₎	59.0	78.2 ₍₃₁₄₎	71.3	89.1 ₍₈₃₇₎	58.7	65.5 ₍₅₈₎	46.4	57.8 ₍₈₆₎
Fr	61.9	80.4 ₍₁₆₀₎	46.4	76.0 ₍₅₀₎	93.4	95.9 ₍₄₀₇₎	7.6	52.1 ₍₁₅₂₎	62.4	66.4 ₍₆₂₎	39.3	61.3 ₍₇₀₎
C3	28.1	71.5 ₍₇₈₎	11.9	57.4 ₍₁₇₎	11.0	54.4 ₍₈₀₎	2.1	42.8 ₍₄₇₎	63.1	75.7 ₍₅₄₎	34.3	66.2 ₍₅₅₎
GL	94.2	96.5 ₍₂₀₄₎	74.7	91.4 ₍₆₇₎	73.7	84.4 ₍₃₆₆₎	82.3	93.0 ₍₉₂₇₎	80.6	85.8 ₍₆₂₎	60.4	68.6 ₍₉₆₎
OL	83.5	90.4 ₍₁₉₃₎	45.4	82.7 ₍₄₅₎	25.1	67.3 ₍₁₅₆₎	3.3	44.6 ₍₇₈₎	83.4	88.7 ₍₆₂₎	47.4	68.0 ₍₇₆₎

Table 3: AfroLID (AL), CLD2 (C2), fasttext (FT), FUN-LangID (FL), pyFranc (Fr), CLD3 (C3), GlotLID (GL), OpenLID-v2 (OL). Scores over whole dataset (*all*) and on subset of covered languages (*cov.*). (x) = Count of languages in the eval. set covered by model.

Performance on Common Subsets of Languages

	High-coverage 1351 languages		Afro* 294 languages		Core 76 languages	
	F1 ↑	FPR ↓	F1 ↑	FPR ↓	F1 ↑	FPR ↓
GL	87.8	0.006%	88.7	0.02%	91.6	0.1%
FL	76.3	0.02%	73.5	0.08%	77.9	0.2%
AL	—	—	71.8	0.7%	—	—
Fr	—	—	—	—	77.1	0.4%
FT	—	—	—	—	77.7	0.5%
OL	—	—	—	—	78.5	0.5%
C2	—	—	—	—	84.7	0.2%
C3	—	—	—	—	61.6	1.4%

Table 4: Performance on combined test data. GlotLID (GL), FUN-LangID (FL), AfroLID (AL), pyFranc (Fr), fasttext (FT), OpenLID-v2 (OL), CLD2 (C2), CLD3 (C3).

Inference Speed Varies Greatly by Model

Model	Samples/s	Duration @1e + 6 samples
CLD2	43735	22s
OpenLID-v2	13123	1m 16s
fasttext	10867	1m 32s
FUN-LangID	7237	2m 18s
GlottLID	3127	5m 19s
gCLD3	‡2026	8m 13s
Franc	520	32m 03s
AfroLID	†66	4h 12m 49s

Table 5: Inference speed for LID models on FLORES+. Inference done on a 14-core Apple M4 Pro chip with 64GB of RAM. †AfroLID uses transformers - performance is likely better on CUDA hardware. ‡Measured on an AMD EPYC 7351P (16 cores @ 2.4GHz) Linux machine with 256GB RAM as gCLD3 not usable on modern macOS.

Trade Off Between Inference Speed and F1

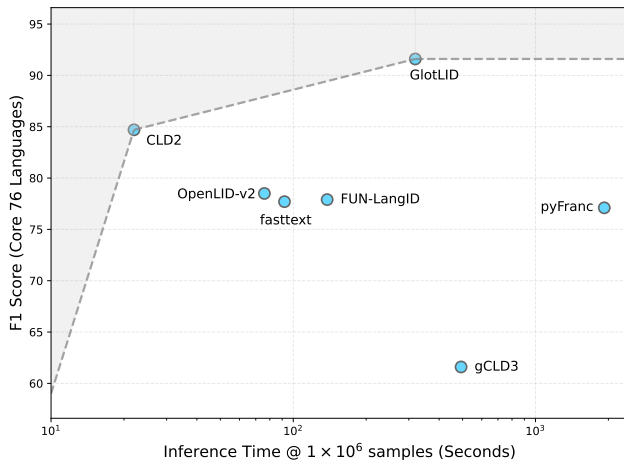


Figure 10: Inference speed vs. F1 (76 core languages, combined datasets)

GlottLID Beats LLM Performance for LID

	High-cov. 1351 langs.		Afro* 294 langs.		Core 76 langs.	
	F1 ↑	FPR ↓	F1 ↑	FPR ↓	F1 ↑	FPR ↓
GlottLID	90.0	0.01%	90.6	0.05%	93.5	0.13%
<i>OpenAI GPT models</i>						
4o-mini	48.5	0.09%	43.6	0.69%	81.0	0.54%
4o	62.0	0.07%	57.3	0.46%	89.0	0.33%
5-mini	60.4	0.06%	55.1	0.43%	76.4	0.53%
5	70.3	0.05%	66.6	0.29%	91.8	0.22%

Table 6: A comparison of large language models from OpenAI’s GPT family (4o-mini, 4o, 5-mini, 5) and GlottLID (best performing LID model). We report performance on the combined but down-sampled test data (15k test samples), subsets as in .

ArXiv Paper



<https://www.arxiv.org/abs/2601.18026>

Common Crawl – A Brief Introduction

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Summary

- Multilinguality is hard!
- Language Identification is hard!
 - Comparing and evaluating models is very difficult
 - Assessing language coverage is complicated, especially when different language registers are considered
 - We need more community initiatives to build better datasets
 - The Common Crawl Foundation will continue working on these language initiatives, but we need your help!
 - We need to develop better architectures and ensure long-term maintenance

Questions?



<https://github.com/commoncrawl/presentations>

References i

- [1] Amazon Web Services. **Open Data Sponsorship Program.**
<https://aws.amazon.com/opendata/open-data-sponsorship-program/>.
- [2] Amazon Web Services. **Registry of Open Data on AWS.**
<https://registry.opendata.aws/>.
- [3] Martijn Koster et al. **Robots Exclusion Protocol.** Tech. rep. 9309. Sept. 2022. 12 pp. <https://www.rfc-editor.org/info/rfc9309>.
- [4] Martijn Koster. **A Standard for Robot Exclusion.** 1996.
<https://www.robotstxt.org/meta.html>.
- [5] Gary Illyes. **Robots Exclusion Protocol Extension for URI Level Control.** Internet-Draft draft-illyes-repext-02. Work in Progress. Internet Engineering Task Force, Oct. 2024. 6 pp.
<https://datatracker.ietf.org/doc/draft-illyes-repext/02/>.

References ii

- [6] Pedro Javier Ortiz Suárez, Laurent Romary, and Benoît Sagot. **“A Monolingual Approach to Contextualized Word Embeddings for Mid-Resource Languages”**. In: *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*. Ed. by Dan Jurafsky et al. Online: Association for Computational Linguistics, July 2020, pp. 1703–1714.
<https://aclanthology.org/2020.acl-main.156/>.
- [7] Julien Abadji et al. **“Towards a Cleaner Document-Oriented Multilingual Crawled Corpus”**. In: *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. Ed. by Nicoletta Calzolari et al. Marseille, France: European Language Resources Association, June 2022, pp. 4344–4355.
<https://aclanthology.org/2022.lrec-1.463/>.

References **iii**

- [8] Kenneth Heafield. “**KenLM: Faster and Smaller Language Model Queries**”. In: *Proceedings of the Sixth Workshop on Statistical Machine Translation*. Ed. by Chris Callison-Burch et al. Edinburgh, Scotland: Association for Computational Linguistics, July 2011, pp. 187–197.
<https://aclanthology.org/W11-2123/>.
- [9] Guillaume Wenzek et al. “**CCNet: Extracting High Quality Monolingual Datasets from Web Crawl Data**”. eng. In: *Proceedings of the Twelfth Language Resources and Evaluation Conference*. Ed. by Nicoletta Calzolari et al. Marseille, France: European Language Resources Association, May 2020, pp. 4003–4012. <https://aclanthology.org/2020.lrec-1.494/>.

References iv

- [10] Linting Xue et al. **“mT5: A Massively Multilingual Pre-trained Text-to-Text Transformer”**. In: *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Ed. by Kristina Toutanova et al. Online: Association for Computational Linguistics, June 2021, pp. 483–498.
<https://aclanthology.org/2021.naacl-main.41/>.
- [11] Guilherme Penedo et al. **“The RefinedWeb dataset for falcon LLM: outperforming curated corpora with web data only”**. In: *Proceedings of the 37th International Conference on Neural Information Processing Systems*. NIPS '23. New Orleans, LA, USA: Curran Associates Inc., 2023.
- [12] Hugging Face. **DataTrove Library**.
<https://github.com/huggingface/datatrove>.

References v

- [13] Luca Soldaini et al. **“Dolma: an Open Corpus of Three Trillion Tokens for Language Model Pretraining Research”**. In: *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Ed. by Lun-Wei Ku, Andre Martins, and Vivek Srikumar. Bangkok, Thailand: Association for Computational Linguistics, Aug. 2024, pp. 15725–15788. <https://aclanthology.org/2024.acl-long.840/>.
- [14] NVIDIA. **NeMo Curator**. <https://github.com/NVIDIA/NeMo-Curator>.
- [15] Ona de Gibert et al. **“A New Massive Multilingual Dataset for High-Performance Language Technologies”**. In: *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*. Ed. by Nicoletta Calzolari et al. Torino, Italia: ELRA and ICCL, May 2024, pp. 1116–1128. <https://aclanthology.org/2024.lrec-main.100/>.

References vi

- [16] Sneha Kudugunta et al. **“MADLAD-400: A Multilingual And Document-Level Large Audited Dataset”**. In: *Advances in Neural Information Processing Systems*. Ed. by A. Oh et al. Vol. 36. Curran Associates, Inc., 2023, pp. 67284–67296.
https://proceedings.neurips.cc/paper_files/paper/2023/file/d49042a5d49818711c401d34172f9900-Paper-Datasets_and_Benchmarks.pdf.
- [17] Amir Hossein Kargaran, François Yvon, and Hinrich Schütze. **“GlottCC: An Open Broad-Coverage CommonCrawl Corpus and Pipeline for Minority Languages”**. In: *arXiv e-prints*, arXiv:2410.23825 (Oct. 2024), arXiv:2410.23825. arXiv: 2410.23825 [cs.CL].

References vii

- [18] Manuel Brack et al. **“Community OSCAR: A Community Effort for Multilingual Web Data”**. In: *Proceedings of the Fourth Workshop on Multilingual Representation Learning (MRL 2024)*. Ed. by Jonne Sälevä and Abraham Owodunni. Miami, Florida, USA: Association for Computational Linguistics, Nov. 2024, pp. 232–235. <https://aclanthology.org/2024.mrl-1.19/>.
- [19] ***Statistics of Common Crawl Monthly Archives***.
<https://commoncrawl.github.io/cc-crawl-statistics/>.
- [20] Julia Kreutzer et al. **“Quality at a Glance: An Audit of Web-Crawled Multilingual Datasets”**. In: *Transactions of the Association for Computational Linguistics* 10 (2022). Ed. by Brian Roark and Ani Nenkova, pp. 50–72. <https://aclanthology.org/2022.tacl-1.4/>.

References viii

- [21] **CLD2Owners/cld2**. 2015. <https://github.com/CLD2Owners/cld2> (visited on 10/03/2025).
- [22] **cld2 package : Ubuntu**. <https://launchpad.net/ubuntu/+source/cld2>.
- [23] **commoncrawl/language-detection-cld2**. 2018.
<https://github.com/commoncrawl/language-detection-cld2>.
- [24] Rami Alrfou. **aboSamoor/polyglot**. 2014.
<https://github.com/aboSamoor/polyglot>.
- [25] **google/cld3**. 2016. <https://github.com/google/cld3>.
- [26] **Language identification · fastText**. 2016. <https://fasttext.cc/index.html>.
- [27] Jonathan Dunn. **jonathandunn/idNet**. 2018.
<https://github.com/jonathandunn/idNet>.

References ix

- [28] saffsd. **saffsd/langid.py**. 2011. <https://github.com/saffsd/langid.py>.
- [29] saffsd. **saffsd/langid.c**. 2014. <https://github.com/saffsd/langid.c> (visited on 10/03/2025).
- [30] Dawid Weiss. **carrotsearch/langid-java**. Nov. 2013.
<https://github.com/carrotsearch/langid-java>.
- [31] **optimaize/language-detector**. 2014.
<https://github.com/optimaize/language-detector>.
- [32] Nakatani Shuyo. **shuyo/language-detection**. 2010.
<https://github.com/shuyo/language-detection>.

References x

- [33] Ife Adebara et al. **“AfroLID: A Neural Language Identification Tool for African Languages”**. In: *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Ed. by Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang. Abu Dhabi, United Arab Emirates: Association for Computational Linguistics, Dec. 2022, pp.1958–1981. <https://aclanthology.org/2022.emnlp-main.128/>.
- [34] NLLB Team et al. **“Scaling neural machine translation to 200 languages”**. In: *Nature* 630.8018 (2024), pp. 841–846. ISSN: 1476-4687. <https://doi.org/10.1038/s41586-024-07335-x>.
- [35] Isaac Caswell. **FunLangID**. https://github.com/google-research/url-nlp/tree/main/fun_langid. 2024.
- [36] Titus Wormer. **pyfranc**. 2023. <https://pypi.org/project/pyfranc/>.

References xi

- [37] Alex Salcianu et al. **cld3: Compact Language Detector v3**. <https://github.com/google/cld3>. Archived; Apache-2.0 License. 2020.
- [38] Amir Hossein Kargaran et al. **“GlottLID: Language Identification for Low-Resource Languages”**. In: *Findings of the Association for Computational Linguistics: EMNLP 2023*. Ed. by Houda Bouamor, Juan Pino, and Kalika Bali. Singapore: Association for Computational Linguistics, Dec. 2023, pp. 6155–6218. <https://aclanthology.org/2023.findings-emnlp.410/>.
- [39] Laurie Burchell et al. **“An Open Dataset and Model for Language Identification”**. In: *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*. Ed. by Anna Rogers, Jordan Boyd-Graber, and Naoaki Okazaki. Toronto, Canada: Association for Computational Linguistics, July 2023, pp. 865–879. <https://aclanthology.org/2023.acl-short.75/>.